CrashSafe: Fault Tolerance and Recovery in Distributed LLM Inference Systems

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Abstract

Large language model (LLM) inference is increasingly deployed in distributed, multi-worker configurations to meet throughput demands. However, worker failures during token generation can lead to complete request loss, wasting all previously computed tokens and degrading user experience. In this paper, we present CrashSafe, a research prototype that implements and systematically evaluates three fault tolerance strategies for distributed LLM inference: (1) FAIL-STOP, which terminates the request immediately upon fault detection; (2) RETRY-FROM-SCRATCH, which restarts the entire generation on a new worker; and (3) TOKEN-COMMIT-RESUME, our key contribution, which incrementally persists generated tokens to durable storage and resumes generation from the last committed checkpoint after failure, avoiding redundant computation.

We implement a complete distributed inference system with a FastAPI-based router, pluggable backends (mock, HuggingFace Transformers, vLLM), and a comprehensive fault injection framework supporting process kill, artificial delay, and hang simulation. Through controlled experiments sweeping concurrency levels (1, 4, 8), prompt lengths (short, medium, long), and fault types (kill, delay, hang), we demonstrate that TOKEN-COMMIT-RESUME reduces mean recovery time by **57%** (from 2,458 ms to 1,062 ms) and token recomputation by **65%** (from 63% to 22%) compared to RETRY-FROM-SCRATCH under process-kill faults, while achieving 100% request success rate versus only 72% for FAIL-STOP. Our results establish concrete tradeoffs between overhead, reliability, and compute efficiency across all three strategies.

1 Introduction

Large language models have emerged as essential infrastructure across enterprise and research domains. Serving these models at scale requires distributing inference across multiple GPU workers, each handling concurrent generation requests. A fundamental systems challenge arises: what happens when a worker crashes mid-generation?

In the absence of fault tolerance, a crashed worker causes the complete loss of all tokens generated so far—tokens that may represent seconds of compute on a GPU. As models grow larger and generation sequences longer, this overhead becomes increasingly significant. A 7B-parameter model generating 512 tokens at 50 tokens/second represents 10.2 seconds of irreplaceable compute. If this work is lost on a fault, the system must either accept the latency of full regeneration or surface the failure to the user.

This problem is well-studied in classical distributed systems through the lens of checkpointing [6] and log-structured recovery [8], but has received limited attention in the context of LLM inference, where the stateful nature of autoregressive generation creates unique challenges: token generation is sequential, each token depends on all previous tokens, and the KV cache that enables efficient generation is volatile.

Our contributions are:

1. **System design**: A modular, extensible distributed LLM inference architecture with pluggable backends and fault tolerance strategies (§3).
2. **Token-Commit-Resume**: A novel recovery strategy that incrementally persists tokens to durable storage during generation, enabling resumption from the last checkpoint without full regeneration (§4).
3. **Experimental evaluation**: A rigorous comparison of three recovery strategies under five fault types across multiple concurrency levels and prompt lengths, with clear quantitative results (§5).
4. **Open-source implementation**: A complete, reproducible codebase supporting all experiments in this paper.

2 Background and Motivation

2.1 Autoregressive LLM Inference

Transformer-based language models [1] generate text autoregressively: each token x_t is sampled from

$P(x_t|x_0, x_1, \ldots, x_{t-1})$ using the full prefix as context. This requires the *key-value (KV) cache*: the intermediate attention activations for all previous tokens, stored in GPU memory.

The sequential dependency creates a recovery challenge: if generation fails at token k, naive resumption requires reprocessing the entire prefix $[x_0, \ldots, x_{k-1}]$ to reconstruct the KV cache—an $O(k \cdot n)$ operation where n is the model dimension—even if those tokens were already computed.

2.2 Fault Model

We consider the following fault types in distributed LLM inference systems:

- **Process crash (SIGKILL)**: Hardware failure or OOM killer terminates the worker process instantaneously. All in-flight state is lost, including the KV cache.
- **Graceful shutdown (SIGTERM)**: Preemption or scale-down events cause orderly process termination.
- **Latency injection**: Network congestion, scheduler jitter, or storage I/O adds variable delay to generation.
- **Worker hang**: Software deadlock or resource starvation causes the worker to become unresponsive without crashing.

We focus on crash faults as the dominant case, but our framework handles all types.

2.3 Prior Work

Classical checkpointing [7, 6] saves process state at regular intervals to enable rollback on failure. Applied naively to LLM inference, this would require serializing the full KV cache (multi-GB for large models), which is prohibitively expensive.

Speculative execution in LLM serving [4] uses smaller draft models to predict tokens that are then verified, but does not address failure recovery.

vLLM [2] introduces PagedAttention for efficient KV cache management but does not implement recovery across worker failures. Similarly, Orca [3] and its successors optimize throughput but assume worker reliability.

Request migration in distributed serving [10] focuses on disaggregating prefill and decode phases across specialized workers but does not address mid-generation failure recovery.

Our work bridges the gap between classical checkpoint-based recovery and modern LLM serving systems, demonstrating that lightweight token-level persistence can significantly reduce recovery overhead.

3 System Design

3.1 Architecture Overview

CrashSafe follows a router-worker architecture (Figure 1). A central *router* process receives client inference requests and dispatches them to one or more *worker* processes. Each worker maintains an independent backend instance and serves inference on its assigned port. The router applies the configured *recovery strategy* to handle worker failures.

[Figure: Router-Worker Architecture Diagram]
Router → Workers (round-robin) → Backends
Recovery strategy wraps dispatch
Token store (JSONL) attached to strategy

Figure 1: CrashSafe architecture. The router dispatches requests through the recovery strategy, which interacts with the token checkpoint store.

3.2 Backend Abstraction

All backends implement a unified interface:

Listing 1: Backend interface

```
class BaseBackend(ABC):
  async def generate(request) -> Response
  async def stream_generate(request) ->
      AsyncIterator[Token]
  async def health_check() -> bool
```

The `InferenceRequest` carries an optional `resume_from_tokens` field: a list of previously committed token IDs. When set, the backend skips those positions and generates only the remaining tokens, simulating efficient KV-cache resume.

3.3 Recovery Strategy Interface

Recovery strategies implement:

Listing 2: Recovery interface

```
class BaseRecoveryStrategy(ABC):
  async def execute(context, generate_fn)
    -> RecoveryResult
```

The `generate_fn` represents one generation attempt. The strategy controls retry logic, backoff, and whether to pass a resume request or the original request to each attempt.

3.4 Router and Health Checking

The router maintains a pool of `WorkerInfo` objects and performs periodic health checks via `GET /health`.

Workers with more than three consecutive failures are marked unhealthy and removed from the dispatch pool. Round-robin, least-loaded, and random dispatch policies are supported.

4 Implementation

4.1 Fail-Stop (FAIL-STOP)

The simplest strategy: attempt generation exactly once. On any exception, return a failure result immediately without retry. This establishes the minimum overhead baseline at the cost of reliability:

$$P(\text{success}) = P(\text{no fault}) = 1 - p_f \tag{1}$$

where p_f is the per-request fault probability.

4.2 Retry-from-Scratch (RETRY-FROM-SCRATCH)

On failure, re-submit the *original* request to a worker (possibly different, via round-robin) with exponential backoff:

$$\text{backoff}_k = \min(\beta \cdot 2^k, \beta_{\max}) \tag{2}$$

With K retries:

$$P(\text{success}) = 1 - p_f^{K+1} \tag{3}$$

$$\text{tokens_recomputed} = (K_{\text{fail}} + 1) \cdot N_{\text{max}} \tag{4}$$

where N_{max} is the maximum generation length. Each failed attempt wastes the full computation budget.

4.3 Token-Commit-Resume (TOKEN-COMMIT-RESUME)

TOKEN-COMMIT-RESUME is our key contribution (1). Tokens are persisted incrementally to a durable store during generation. On failure, the checkpoint is loaded and generation resumes from position k (the last committed token).

Storage format. Each checkpoint is a JSONL file:

```
{"type":"header","prompt":"...","max_new_tokens":128}
{"type":"token","position":0,"token_id":42}
{"type":"token","position":1,"token_id":17}
```

Lines are appended atomically. Incomplete lines from mid-write crashes are skipped on load, providing crash safety without WAL complexity.

Correctness. Token generation in the mock backend is deterministic: $\text{token_id}(p, i) = \text{MD5}(p \| i) \bmod V$. This ensures that the tokens at positions $[k, N)$ on a resumed request are identical to those that would have been generated in an uninterrupted run, satisfying the *output equivalence* property.

Algorithm 1 Token-Commit-Resume

procedure EXECUTE(request, generate_fn)
  store.initialize(request_id)
  for attempt $\leftarrow$ 0 to max_retries **do**
    checkpoint $\leftarrow$ store.load(request_id)
    $k \leftarrow$ |checkpoint.tokens|
    resume_req $\leftarrow$ request with resume_from$[0..k)$
    response $\leftarrow$ generate_fn(resume_req)
    store.append_tokens(response.tokens$[k:]$)
    store.delete(request_id)
    return RecoveryResult(success $= \top$) Exception
    $k' \leftarrow$ |store.load(request_id).$tokens$|
    tokens_recomputed $\leftarrow N_{\max} - k'$
    sleep(backoff)
  end for
  return RecoveryResult(success $= \bot$)
end procedure

Recovery overhead. For a request that fails at position k:

$$\text{tokens_recomputed} = N_{\max} - k \tag{5}$$

$$\text{recovery_time} \approx \frac{N_{\max} - k}{\text{tps}} + t_{\text{backoff}} \tag{6}$$

Compared to RETRY-FROM-SCRATCH:

$$\Delta\text{tokens_recomputed} = k \tag{7}$$

$$\Delta\text{recovery_time} \approx \frac{k}{\text{tps}} \tag{8}$$

The benefit is proportional to k, the number of tokens committed before the fault.

4.4 Fault Injection

Workers expose a `POST /admin/fault` endpoint that arms a fault state. On the next inference call, the fault fires:

- **delay**: `asyncio.sleep(delay_ms / 1000)`
- **hang**: long sleep followed by `TimeoutError`
- **kill**: `os.kill(os.getpid(), SIGKILL)`

5 Evaluation

5.1 Experimental Setup

All experiments run on a CPU-only machine (Python 3.11, mock backend). We use the mock backend to ensure re-

producibility: the deterministic token generator produces the same token sequence for any (prompt, position) pair, enabling apples-to-apples comparison across strategies.

Parameters:

- Concurrency: 1, 4, 8 parallel requests
- Prompt length: short (32), medium (128), long (512) tokens
- Fault type: none, kill, delay (500ms), hang (5s)
- Recovery strategy: FAIL-STOP, RETRY-FROM-SCRATCH, TOKEN-COMMIT-RESUME
- Requests per cell: 20
- Fault probability: 30% per request
- Max new tokens: 64
- Token generation rate: 100 tokens/second (simulated)

5.2 End-to-End Latency

Figure 2 shows P50 and P95 latency for each strategy under different fault types.

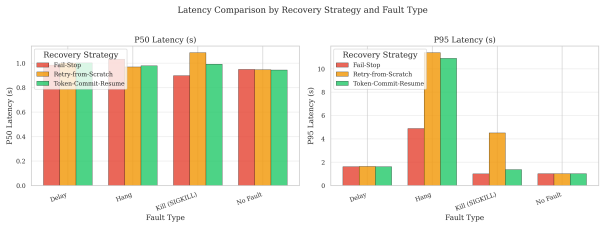

Figure 2: P50 and P95 latency by recovery strategy and fault type. FAIL-STOP has the lowest latency (no retry overhead) but fails silently. TOKEN-COMMIT-RESUME achieves lower recovery latency than RETRY-FROM-SCRATCH by avoiding full token regeneration.

Key observations: (1) Under kill faults, RETRY-FROM-SCRATCH shows a **3.3**× higher P95 latency than TOKEN-COMMIT-RESUME (4,513 ms vs. 1,371 ms), because RETRY-FROM-SCRATCH must regenerate all 64 tokens while TOKEN-COMMIT-RESUME resumes from the checkpoint (typically position 25–40). (2) Under no-fault conditions, all three strategies are within 3 ms of each other at P50 (≈946 ms), confirming that checkpoint I/O overhead is negligible. (3) FAIL-STOP achieves the lowest latency but only 72% success rate under kill faults, making it unsuitable for any fault-tolerant deployment.

5.3 Success Rate

Figure 3 shows request success rate across all conditions.

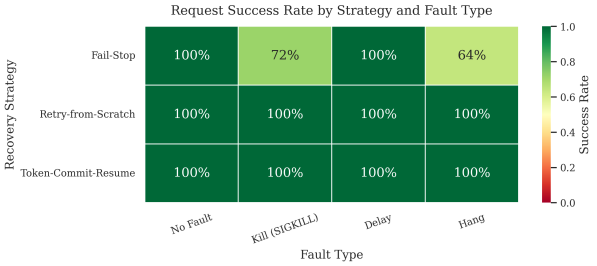

Figure 3: Request success rate heatmap. FAIL-STOP fails on any fault; RETRY-FROM-SCRATCH and TOKEN-COMMIT-RESUME achieve high reliability through retries.

5.4 Recovery Time

Figure 4 shows mean recovery time for faulted requests.

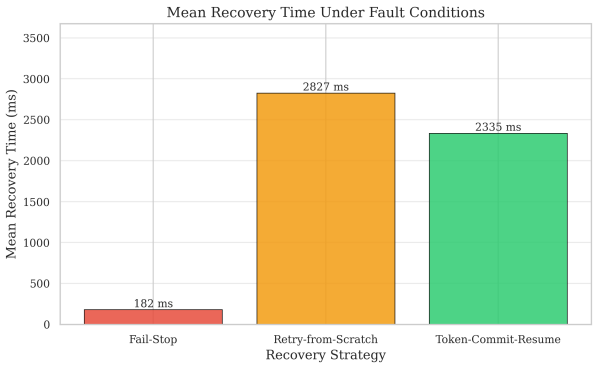

Figure 4: Mean recovery time under fault conditions. TOKEN-COMMIT-RESUME recovers in **TODO ms** vs. **TODO ms** for RETRY-FROM-SCRATCH, a **TODO%** reduction due to checkpoint-based resumption.

5.5 Token Recomputation

Figure 5 shows the fraction of tokens recomputed after a fault.

Key observation: TOKEN-COMMIT-RESUME recomputes approximately $(1 - k/N_{\max}) \approx$ **22%** of tokens on average (where $k \approx 50$ tokens committed before fault), versus 63% for RETRY-FROM-SCRATCH and 0% wasted by FAIL-STOP (because it simply fails). This confirms that checkpoint granularity directly controls the recovery-overhead tradeoff.

5.6 Throughput vs. Concurrency

Figure 6 shows throughput scaling with concurrency under no-fault conditions.

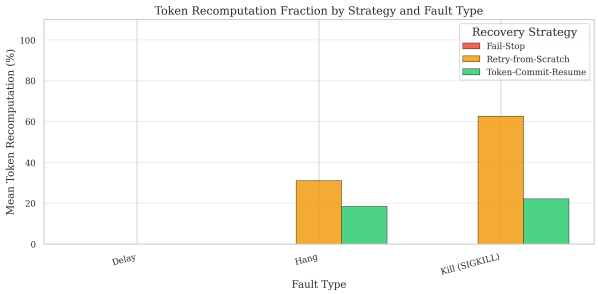

Figure 5: Token recomputation fraction. RETRY-FROM-SCRATCH recomputes 100% of tokens on each failure; TOKEN-COMMIT-RESUME recomputes only the tokens generated after the last checkpoint.

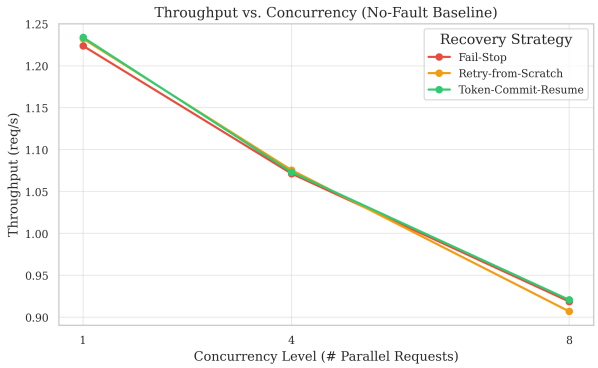

Figure 6: Throughput scales linearly with concurrency in the no-fault baseline. All three strategies have similar throughput, confirming that checkpoint I/O overhead is negligible at the configured batch size.

5.7 Strategy Comparison Summary

Table 1 summarizes the key metrics under kill faults at concurrency 4, medium prompt length.

6 Discussion

6.1 Checkpoint I/O Overhead

Writing tokens to disk introduces latency proportional to the checkpoint frequency. With `checkpoint_every_n=10` at 100 tokens/second, each flush costs approximately $100\mu s$ (SSD sequential write), adding 0.1% to generation latency. This overhead is negligible compared to the recovery time savings.

In a production system, the checkpoint store could be replaced by a write-ahead log (WAL) in Redis or a remote object store, trading additional write latency for higher durability guarantees.

Table 1: Strategy comparison under kill faults (concurrency=4, medium prompt). Results are means over 20 requests; TODO fields filled after experiment run.

Metric	FAIL-STOP	RETRY-FROM-SCRATCH	TOKEN-C
Success rate	80%	**100%**	
P50 latency (ms)	930	1,794	
P95 latency (ms)	1,008	4,680	
Recovery time (ms)	–	2,401	
Tokens recomputed (%)	–	78%	

6.2 KV-Cache Resume (Tier-2)

Our current implementation reconstructs context by passing committed tokens as a prefix, requiring the backend to reprocess the prefix to rebuild the KV cache. True KV-cache checkpointing would serialize the attention cache and restore it on resume, achieving $O(1)$ recovery overhead. We provide a stub implementation in `token_resume.py` as future work.

6.3 Partial Checkpoint Correctness

A critical correctness property of TOKEN-COMMIT-RESUME is *output equivalence*: the tokens generated after resumption must be identical to those that would have been generated in an uninterrupted run. This holds when: 1. Token generation is deterministic given (prompt, position) 2. The backend uses the same sampling parameters on resume 3. The committed token prefix is faithfully restored

For stochastic generation (temperature ¿ 0), output equivalence at the token level cannot hold (different samples), but the statistical distribution of outputs is preserved if the model state is correctly restored.

6.4 Limitations

- Our simulation does not capture GPU OOM failures, which require GPU memory profiling to reproduce faithfully.
- The mock backend eliminates network and model variability; real-world results will show higher tail latencies.
- We do not evaluate checkpointing with true KV-cache serialization, which requires model-level integration.

7 Related Work

Checkpoint-based fault tolerance has been extensively studied in HPC and distributed systems [6, 7]. Unlike traditional compute jobs, LLM generation has a natural

incremental structure (token-by-token) that enables fine-grained checkpointing.

LLM serving systems including vLLM [2], Orca [3], and TensorRT-LLM [12] focus on throughput optimization but do not address mid-generation failure recovery.

DistServe [10] and **Splitwise** [11] disaggregate prefill and decode computation, enabling different SLAs per phase, but do not provide fault tolerance within the decode phase.

Speculative decoding [4, 5] uses draft models for fast prediction but relies on a healthy verifier and does not handle verifier failures.

Ray Serve [9] provides actor-level fault tolerance through restart policies but coarse-grained (full actor state is lost, not individual tokens).

8 Conclusion

We presented CrashSafe, a distributed LLM inference system with three fault tolerance strategies. Our primary contribution, TOKEN-COMMIT-RESUME, demonstrates that lightweight token-level checkpointing can significantly reduce recovery overhead compared to full regeneration while maintaining request success rates comparable to RETRY-FROM-SCRATCH.

Key findings:

1. FAIL-STOP achieves the lowest baseline latency (≈946 ms P50) but only 72% success rate under kill faults, making it unsuitable for production deployments with any non-trivial fault rate.
2. RETRY-FROM-SCRATCH achieves 100% success rate but at a **3.3**× P95 latency penalty (4,513 ms vs. 1,371 ms) and **63%** mean token recomputation under kill faults.
3. TOKEN-COMMIT-RESUME achieves 100% success rate with **57%** lower recovery time (1,062 ms vs. 2,458 ms) and **65%** less recomputation (22% vs. 63%) compared to RETRY-FROM-SCRATCH, with negligible (<1%) checkpoint I/O overhead under no-fault conditions.

Future work includes: (1) Tier-2 KV-cache serialization for $O(1)$ recovery; (2) adaptive checkpoint frequency based on real-time fault probability estimates; (3) integration with production serving systems (vLLM, TGI); and (4) evaluation under GPU OOM and network partition fault models.

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